Status	Finished
Started	Thursday, 29 October 2026, 11:02 PM
Completed	Thursday, 29 October 2026, 11:05 PM
Duration	3 mins 19 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O👎 ✓
- b. O(n log n)
- c. O(log n)
- d. O(1)

Question **2**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

- a. Bubble ✓
- b. Insertion
- c. Selection
- d. Complexity

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Sorting the list before searching
- b. Checking each element sequentially ✓
- c. Dividing the list in half repeatedly
- d. Skipping every second element

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is very small
b. When the list is unsorted
c. When the list contains duplicate elements
d. When the list is sorted ✓

Question **5**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Hashing
b. Jump search
c. Linear search ✓
d. Binary search

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search requires the list to be sorted.
b. Linear search can be applied to both sorted and unsorted lists. ✓
c. Linear search divides the list into halves.
d. Linear search is more efficient than binary search on large lists.

Question **7**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It can be applied only if the list is sorted ✓
b. It works on unsorted lists
c. It compares elements sequentially
d. It always starts from the beginning of the list

Question **8**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list is sorted
b. When the element is found in the list ✓
c. When the list contains duplicate elements
d. When the search algorithm finishes

Question **9**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Hash search
b. Linear search
c. Binary search ✓
d. Sequential search

Question **10**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the right sublist
b. In the left sublist ✓
c. In the entire list
d. At the beginning of the list

Question **11**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Finding
b. Mining
c. Discovering
d. Searching ✓

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Binary search
b. Hashing
c. Linear search ✓
d. Dynamic search

Question **13**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search can find multiple instances of the target element
b. Binary search does not require dividing the list
c. Binary search has a lower time complexity on large, sorted lists ✓
d. Binary search works on unsorted lists

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It is slower than linear search for small lists
b. It does not work with negative numbers
c. It requires the list to be sorted ✓
d. It can only be applied to large lists

Question **15**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 100
b. 94 and 99
c. 90 and 99 ✓
d. 89 and 94